

```
In [155... import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
```

```
In [157... data = pd.read_excel("data/distance_vs_time.xlsx")
```

```
In [159... data.head()
```

```
Out[159...      time_s  distance_cm  dist_uncertainty  time_uncertainty
0      0.6         59         15             0.05
1      1.2        132         15             0.05
2      1.8        191         15             0.05
3      2.4        235         15             0.05
```

The Beats per minute used is 100 and we used 4 koosh balls to mark the distance traveled by the car at a consecutive beat

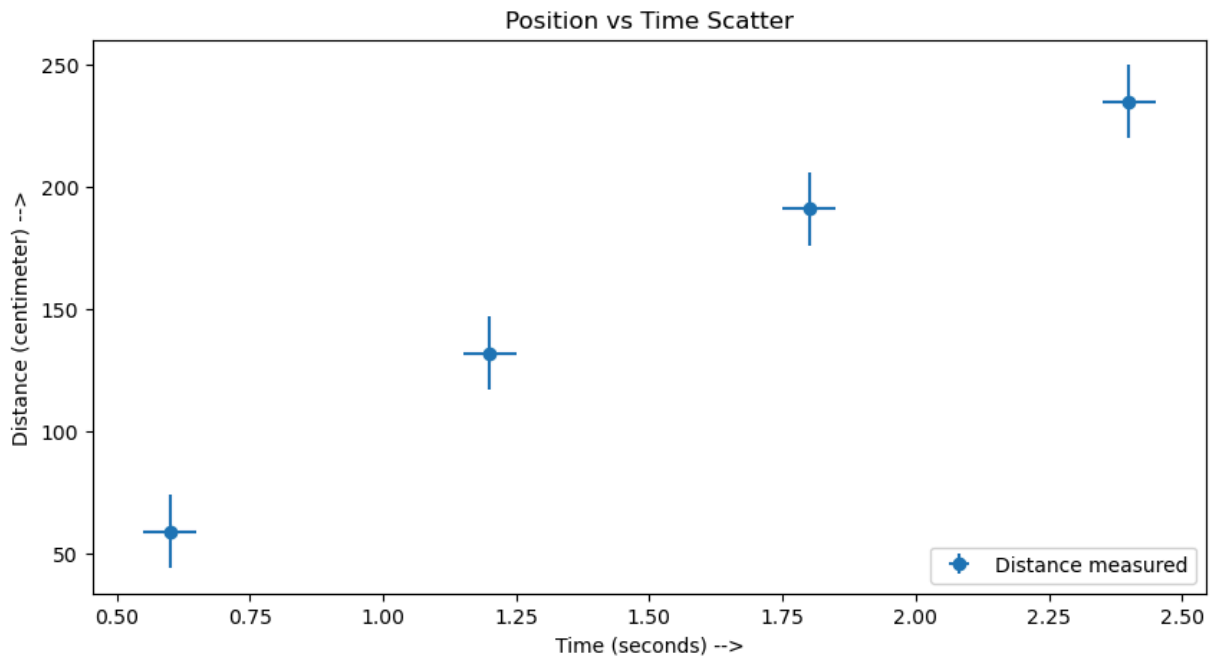
Error Analysis

The average reaction time of a human when for an expected event is around 0.15 sec. Since the koosh balls were dropped at estimated places and not the exact, the uncertainty is around 10 centimeter. The truck also has some uncertainties as it isn't a point mass but deviate around 5 cm from the center.

```
In [164... tot_dist_uncertainty = 5 + 10
tot_time_uncertainty = 0.05
time_unit = "(seconds)"
distance_unit = "(centimeter)"
```

```
In [166... fig, ax = plt.subplots(figsize=(10, 5))
ax.errorbar(data['time_s'], data['distance_cm'], yerr=tot_dist_uncertainty, xerr=t
ax.set_xlabel(f"Time {time_unit} -->")
ax.set_ylabel(f"Distance {distance_unit} -->")
ax.legend(loc='lower right')
ax.set_title("Position vs Time Scatter")
```

```
Out[166... Text(0.5, 1.0, 'Position vs Time Scatter')
```



Generalizing the relationship between the position and time

To get a numerical value for velocity, we fit a straight line to each of the data sets. We use a linear model

$$y = A + Bx$$

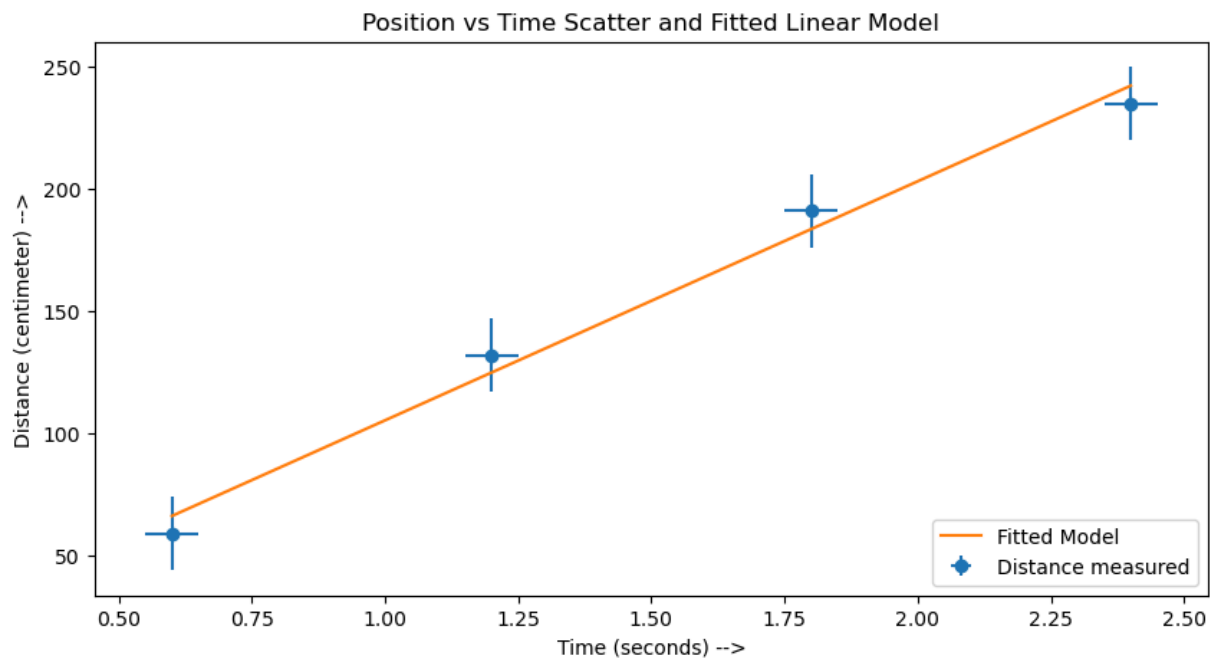
```
In [170...] linear_model = lambda x, A, B: A*x + B
```

```
In [172...] para, cov = curve_fit(linear_model, # linear model
                               data["time_s"], # x-data
                               data["distance_cm"], # y-data)
```

```
In [223...] initial_pos = para[1]
            velocity = para[0]
```

```
In [193...] fig, ax = plt.subplots(figsize=(10, 5))
            ax.errorbar(data['time_s'], data['distance_cm'], yerr=tot_dist_uncertainty, xerr=t
            ax.set_xlabel(f"Time {time_unit} -->")
            ax.set_ylabel(f"Distance {distance_unit} -->")
            ax.set_title("Position vs Time Scatter and Fitted Linear Model")
            time = np.linspace(min(data['time_s']), max(data['time_s']), 1000)
            pos = linear_model(time, velocity, initial_pos)
            ax.plot(time, pos, label="Fitted Model")
            ax.legend(loc='lower right')
```

```
Out[193...] <matplotlib.legend.Legend at 0x19595f0fc90>
```



```
In [228... print(f"Velocity = {velocity}")
print(f"Velocity corrected to the right sig figs = {velocity:.1g}")
```

Velocity = 97.83333330385928

Velocity corrected to the right sig figs = 1e+02

Since our estimated velocity is 97.833-> cm/s but since our uncertainty is 15 cm we can round it to the correct number of significant digits.

$$v_{\text{Velocity}} = 100 \pm 15(\text{cm/s})$$

In []: